Enterprise NeXus — Multi-Agent AI Business Automation OS with Security Intelligence

CSE 4204 — Mobile Computing Lab | Team CSE4204-8B-T04

Week 09 Deliverable — Feature Completion Report

1. Introduction

This report documents the Week 09 milestone of Enterprise NeXus — Multi-Agent AI Business Automation OS with Security Intelligence. Building on the AI integration delivered in Week 08, Week 09 focuses on completing all major features, improving the user experience, optimising performance, and removing outstanding bugs so that the system reaches a near-complete, demonstrable state ahead of formal testing.

2. Team Information

Name	Student ID	Role
Nazmus Sakib	11220320888	Team Leader & AI Integration Lead
Shoeb Shariar Mashuk	11220320878	Backend Developer
Most Sumiya Sanjida	11220320874	Frontend Developer — 1
Sabrina Ibrahim	11220320895	Database Manager & Frontend Developer — 2

3. Feature Completion Status

All core modules promised in the proposal are now implemented end-to-end — backend API, database persistence, AI analysis, and a connected frontend page. The table below summarises completion by module.

Module	Key Capabilities	Status
Authentication	Register, login, JWT refresh, logout, forgot/reset password, RBAC	Complete
HR	CV upload & AI screening, candidate list/detail, stats, HR agent chat	Complete
Finance	Expense records, invoice & bulk upload, anomaly detection, finance agent	Complete
Support	Tickets, sentiment/urgency analysis, escalate/resolve, RAG replies	Complete
Analytics	KPI snapshot, AI reports, charts	Complete
Executive	Multi-agent orchestration, daily briefing, PDF export, chat	Complete

Module	Key Capabilities	Status
Security Intelligence	Audit log viewer, security stats, threat monitoring	Complete
Knowledge Base / RAG	Document upload, embeddings, semantic search	Complete
Admin	User & role management, knowledge management	Complete
Notifications	In-app notifications, preferences, realtime updates	Complete

4. System Checklist

Per the handbook Week 09 checklist, the following were verified across the running system:

- Authentication works — registration, login, token refresh, protected routes, and password reset all function correctly.
- APIs work properly — every documented REST endpoint returns correct, validated responses.
- AI responds correctly — all five agents return grounded, schema-valid output and degrade gracefully when the AI is unavailable.
- Database stores data correctly — records persist with Row-Level Security enforcing per-user isolation.
- UI is responsive — dashboard, forms, and AI screens render on desktop, tablet, and mobile.
- No broken navigation — sidebar, top bar, and all detail routes resolve; a custom 404 page handles unknown routes.

5. UI/UX Improvements

The Week 09 iteration refined the interface delivered in Week 07 for clarity and polish:

- Loading, empty, and error states standardised through reusable Skeleton, EmptyState, and ErrorBoundary components.
- Consistent StatCard, Badge, Pagination, and PageHeader components across all modules.
- A first-run WelcomeTour to onboard new users, and toast-based feedback for all actions.
- Real-time dashboard refresh and live agent-activity feed via Supabase Realtime.
- Colour-coded severity, sentiment, and recommendation badges for fast scanning.

6. Performance Optimisation

- SWR (stale-while-revalidate) caching on the frontend to avoid redundant fetches.
- Database indexes on all foreign keys and frequently-queried columns (status, recommendation, severity, dates).
- Response compression and pagination on list endpoints to keep payloads small.

- Bounded AI tool loops (max five iterations) and a per-call timeout to keep agent latency predictable.

7. Bugs Resolved This Week

Area	Issue	Resolution
AI output	Occasional non-JSON model responses	Switched to Gemini native JSON mode with response schema
Auth	Refresh token not revoked on logout	Hashed refresh tokens stored in user_sessions and invalidated
Finance	Bulk upload rejected valid spreadsheets	Hardened file parsing and validation
Support	Replies not grounded in policy	Wired retrieve_knowledge RAG tool into draft replies
UI	Layout breakpoints on tablet	Adjusted responsive grid and sidebar behaviour

8. Deliverables Summary

- Nearly complete project — all modules implemented, integrated, and demonstrable.
- Feature checklist — the verified system checklist in Section 4.
- Updated GitHub repository reflecting Week 09 progress.